

Completely bounded multipliers of a Beurling–Fourier algebra form a dual Banach algebra

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Abstract

Alaghmandan, Giselsson, Samei, and Turowska prove that, for every locally compact group G and weight inverse ω , the completely bounded multiplier algebra $M_{\text{cb}}A(G, \omega)$ is completely isometric to the dual of the balanced operator-projective tensor product $A(G, \omega) \widehat{\otimes}_{A(G, \omega)} C_r^*(G)$. They state that they believe this multiplier algebra is a dual Banach algebra, proving that fact only when G is amenable. We prove the general statement. For a fixed multiplier ψ , its multiplication map on $A(G, \omega)$ preserves the balanced tensor relations and therefore induces a completely bounded map S_ψ on the displayed predual. The adjoint S_ψ^* is exactly multiplication by ψ on $M_{\text{cb}}A(G, \omega)$. Multiplication is consequently separately weak-star continuous for the source predual.

1 Source statement

The source is M. Alaghmandan, O. Giselsson, E. Samei, and L. Turowska, *Multipliers of Beurling–Fourier algebras*, arXiv:2606.06675v1 [1]. On page 27, immediately before Theorem 7.12, the authors write:

“It is known that $M_{\text{cb}}A(G)$ is a dual Banach algebra. We believe that this should hold for $M_{\text{cb}}A(G, \omega)$ as well. By Theorem 8.3 and Lemma 5.5, this is true for amenable groups.”

The full page-width source passage, including the balanced tensor definition, the conjectural sentence, and Theorem 7.12, is reproduced below.

Let A be a completely contractive Banach algebra, X is right A -module and Y is a left A -module. The A -module tensor product of X and Y is the quotient space $X \hat{\otimes}_A Y := X \otimes Y / N$, where

$$N = \langle x \cdot a \otimes y - x \otimes a \cdot y \mid x \in X, y \in Y, a \in A \rangle,$$

and $\langle \cdot \rangle$ denotes the closed linear span. It follows that

$$CB_A(X, Y^*) \cong N^\perp \cong (X \hat{\otimes}_A Y)^*,$$

where $CB_A(X, Y^*)$ denotes the space of completely bounded right A -module maps $\Phi : X \rightarrow Y^*$ (see [3, Corollary 3.5.10]).

In [9, Remark 5.9], J. Crann showed that, for a locally compact group G , $M_{cb}A(G)$ is completely isometric to the dual space of $A(G) \hat{\otimes}_{A(G)} C_r^*(G)$. In the theorem below, we extend this result to Beurling-Fourier algebras and show that $M_{cb}A(G, \omega)$ is a dual operator space. It is known that $M_{cb}A(G)$ is a dual Banach algebra. We believe that this should hold for $M_{cb}A(G, \omega)$ as well. By Theorem 8.3 and Lemma 5.5, this is true for amenable groups.

Theorem 7.12 *Let G be a locally compact with a weight inverse ω . Then $M_{cb}A(G, \omega)$ is completely isometric to the dual space of $A(G, \omega) \hat{\otimes}_{A(G, \omega)} C_r^*(G)$.*

Proof. Consider $A(G, \omega) \hat{\otimes}_{A(G, \omega)} C_r^*(G)$. Then it is known that $(A(G, \omega) \hat{\otimes}_{A(G, \omega)} C_r^*(G))^*$ is isometrically isomorphic to $CB(A(G, \omega), B_r(G, \omega))$ by the equation [13, 7.1.15]. Now let X be the closed linear subspace of $A(G, \omega) \hat{\otimes}_{A(G, \omega)} C_r^*(G)$ given by

$$X := \langle (\omega u)(\omega v) \otimes x - (\omega u \otimes (\omega v)) \cdot_\omega x : u, v \in A(G, \omega), x \in C_r^*(G) \rangle.$$

Then $X^\perp \subseteq CB(A(G, \omega), B_r(G, \omega))$ is a weak*-closed subspace. Also, for each T in $CB(A(G, \omega), B_r(G, \omega))$ we have that $T \in X^\perp$ if and only if $T((\omega u)(\omega v)) = T(\omega u)(\omega v)$. But, by Lemma 7.11, we know that products are dense in $A(G, \omega)$ and, by Theorem 5.3, $A(G, \omega)$ is an ideal in $B_r(G, \omega)$. Therefore, $T(A(G, \omega)) \subseteq A(G, \omega)$. Consequently, there is some $\phi \in M_{cb}A(G, \omega)$ so that $\mathbf{m}_\phi = T$. We have thus shown that $M_{cb}A(G, \omega)$ is completely isometric to $X^\perp$, and hence $M_{cb}A(G, \omega)$ is a dual Banach space where the weak* topology on $M_{cb}A(G, \omega)$ is derived through embedding $M_{cb}A(G, \omega) \rightarrow (A(G, \omega) \hat{\otimes}_{A(G, \omega)} C_r^*(G))^*$ given by

$$\langle \mathbf{m}_\phi, (\omega u) \otimes x \rangle = \langle \mathbf{m}_\phi(\omega u), x \rangle_\omega.$$

□

Theorem 7.12 of the source proves, for arbitrary locally compact G and every weight inverse ω , the completely isometric identification

$$M_{cb}A(G, \omega) \cong (A(G, \omega) \hat{\otimes}_{A(G, \omega)} C_r^*(G))^*.$$

The unresolved point is whether multiplication is separately weak-star continuous for this predual.

2 Proof intuition

The predual constructed by the source already remembers the multiplier module structure. Indeed, let ψ be a fixed completely bounded multiplier. Applying ψ to the first tensor factor sends

$$ab \otimes x - a \otimes b \cdot x \quad \text{to} \quad \psi(ab) \otimes x - (\psi a) \otimes b \cdot x.$$

The multiplier identity $\psi(ab) = (\psi a)b$ says that the image is another balanced relation. Thus the map descends to the quotient predual. Under the source pairing, its adjoint sends a multiplier ϕ to the product $\phi\psi$. This gives the missing preadjoint of multiplication directly; no approximation, amenability, or additional property of the weight is needed.

3 An abstract tensor lemma

We use the operator-space projective tensor product and its standard functoriality [2]. The proof below also records explicitly the only quotient issue that matters.

Lemma 1. *Let $\mathcal{A}$ be a commutative completely contractive Banach algebra, let X be a left operator $\mathcal{A}$ -module, and put*

$$P = \mathcal{A} \widehat{\otimes}_{\mathcal{A}} X.$$

Suppose $M_{\text{cb}}(\mathcal{A})$ is identified with P^ through the canonical pairing*

$$\langle \phi, [a \otimes x] \rangle = \langle \phi(a), x \rangle.$$

Then this identification makes $M_{\text{cb}}(\mathcal{A})$ a dual Banach algebra. More precisely, for every $\psi \in M_{\text{cb}}(\mathcal{A})$, multiplication by ψ is the adjoint of a completely bounded operator on P .

Proof. Write $R_\psi : \mathcal{A} \rightarrow \mathcal{A}$ for the completely bounded multiplier map $R_\psi(a) = \psi(a)$. By functoriality, $R_\psi \widehat{\otimes} \text{id}_X$ is completely bounded on $\mathcal{A} \widehat{\otimes} X$.

Let N be the closed balanced subspace generated by

$$ab \otimes x - a \otimes b \cdot x \quad (a, b \in \mathcal{A}, x \in X).$$

Since R_ψ is a right $\mathcal{A}$ -module map,

$$R_\psi(ab) = R_\psi(a)b.$$

Consequently

$$\begin{aligned} (R_\psi \widehat{\otimes} \text{id}_X)(ab \otimes x - a \otimes b \cdot x) &= R_\psi(ab) \otimes x - R_\psi(a) \otimes b \cdot x \\ &= (R_\psi(a)b) \otimes x - R_\psi(a) \otimes b \cdot x \in N. \end{aligned}$$

Thus $R_\psi \widehat{\otimes} \text{id}_X$ descends to a completely bounded operator $S_\psi : P \rightarrow P$ satisfying

$$S_\psi[a \otimes x] = [\psi(a) \otimes x].$$

For $\phi \in M_{\text{cb}}(\mathcal{A})$ and an elementary balanced tensor,

$$\begin{aligned} \langle S_\psi^* \phi, [a \otimes x] \rangle &= \langle \phi, S_\psi[a \otimes x] \rangle \\ &= \langle \phi(\psi(a)), x \rangle \\ &= \langle (\phi\psi)(a), x \rangle \\ &= \langle \phi\psi, [a \otimes x] \rangle. \end{aligned}$$

Elementary tensors are dense in P , so $S_\psi^* \phi = \phi\psi$. Hence multiplication by each fixed ψ is weak-star continuous. Since $\mathcal{A}$ is commutative, this proves separate weak-star continuity of multiplication. $\square$

4 Resolution of the source conjecture

Theorem 2. *Let G be a locally compact group and let ω be a weight inverse on G . With the predual from Theorem 7.12 of the source,*

$$M_{\text{cb}}A(G, \omega) = (A(G, \omega) \widehat{\otimes}_{A(G, \omega)} C_r^*(G))^*$$

is a dual Banach algebra. In fact, multiplication is separately completely weak-star continuous.

Proof. Set $\mathcal{A} = A(G, \omega)$ and $X = C_r^*(G)$, with the left $\mathcal{A}$ -module action used in the source balanced tensor product. The algebra $\mathcal{A}$ is a commutative completely contractive Banach algebra, and a member of $M_{\text{cb}}A(G, \omega)$ is precisely a completely bounded multiplier of $\mathcal{A}$. Theorem 7.12 of the source supplies the completely isometric dual identification and the canonical pairing required in Lemma 1. Applying that lemma proves the assertion. The maps S_ψ are completely bounded, so their adjoints are completely weak-star continuous at every matrix level. $\square$

5 Verification and novelty notes

The argument depends only on the source’s Theorem 7.12 and on functoriality of the operator-projective tensor product. The balanced quotient is checked in line. The key identity can be audited on one elementary tensor:

$$\langle \phi\psi, [a \otimes x] \rangle = \langle \phi, [\psi(a) \otimes x] \rangle.$$

This identifies multiplication with an adjoint operator before taking any limit, so no density-of-products argument or uniform-boundedness step is needed.

A bounded novelty check on August 11, 2026 searched all four lightweight run indexes for arXiv:2606.06675, Beurling–Fourier multipliers, and dual Banach algebras. External searches covered the exact title and arXiv id and the phrases “ $M_{\text{cb}}A(G, \omega)$ dual Banach algebra”, “Beurling–Fourier dual Banach algebra”, and “balanced tensor multiplier predual”. The arXiv record still has only v1, submitted June 4, 2026. No separate answer or matching general theorem was found. Novelty remains subject to specialist review.

6 Limitations and human-review recommendation

The result uses exactly the weak-star topology induced by the source predual; it does not assert uniqueness of that predual. It assumes Theorem 7.12 of the source as stated. Within that framework, there is no remaining conditional step and no restriction to amenable groups.

This packet is recommended for expert review as a likely valid full solution. The most important checks are: (i) that the first tensor factor in the source balanced product is the right regular $A(G, \omega)$ -module, (ii) that a completely bounded multiplier is a completely bounded right module map, and (iii) that the source pairing agrees with the pairing used in Lemma 1.

References

- [1] M. Alaghmandan, O. Giselsson, E. Samei, and L. Turowska, *Multipliers of Beurling–Fourier algebras*, arXiv:2606.06675v1, 2026.
- [2] D. P. Blecher and C. Le Merdy, *Operator Algebras and Their Modules: An Operator Space Approach*, Oxford University Press, 2004.